

Task for Tech Cell Coordinator Selection - OCS Report

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CS25BTECH11064

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1 Technology Stack

- **Frontend:** HTML, CSS, JavaScript, Bootstrap 5
- **Backend:** Supabase (handles authentication and database)
- **Database:** PostgreSQL (via Supabase)
- **Hosting:** GitHub Pages (frontend) + Supabase Cloud (backend)

Supabase was chosen because it provides authentication, a database, and a REST API in one platform. No custom backend server was needed, which allowed faster development.

2 Database Design

Three tables were created in the database:

1. **profiles** – stores user name, email, role (admin/user/viewer), and active status
2. **rooms** – stores block name, room name, capacity, availability, and allowed purposes
3. **bookings** – stores date, time, purpose, participants, room, user, and booking status

3 Features Implemented

3.1 Authentication

- Only admin-created accounts can log in

- No double-booking of rooms
- Room capacity is respected
- Admin has full control over users, rooms, and bookings
- No self-registration. Only admin can create user accounts.
- Login uses email and password via Supabase Auth.
- Passwords are hashed and never stored in plain text.
- Role-based redirection: admin goes to admin page, users go to booking page.

3.2 Room Management

- 55 rooms pre-loaded from the IITH room list across all campus blocks.
- Admin can add new rooms, edit existing rooms, or disable rooms.
- Each room has block name, capacity, availability status, allowed purposes and a note.

3.3 Booking

- Users search rooms by date, time, purpose, and number of participants.
- System shows only rooms that are available and match the criteria.
- Double-booking is prevented by a conflict detection query.
- Users can cancel their own bookings.
- Admin can cancel any booking.

3.4 Admin Dashboard

- View total users, rooms, active bookings, and available rooms.
- Create and deactivate user accounts.
- Add, edit, and disable rooms.
- View and cancel all bookings.

4 Conflict Detection Logic

Before confirming a booking, the system checks if the room is already booked during the requested time:

Two bookings overlap when an existing booking starts before the new one ends AND ends after the new one starts. This catches all overlap cases including partial overlaps.

5 Validation Rules

- End time must be after start time
- Number of participants must be at least 1
- Room capacity must be greater than or equal to participants
- Purpose must be one of OA, Interview, or PPT
- No overlapping bookings for the same room on the same date
- Only authenticated users can make bookings

All these were validated properly.

6 Assumptions

- Each user has one role only – admin, user, or viewer
- Bookings are confirmed immediately without an approval step
- Cancelled bookings are kept in the database for record-keeping
- The 55 rooms from the task document are loaded as initial data

7 How to Run

<https://vmss2727.github.io/Task-for-Tech-Cell-Coordinator-Selection-OCS-IITH-May-2026/>

Launch the above url in your browser, you will be redirected to the site.

Admin Credentials are :

Email : admin@ocs.iith.ac.in

Password : 1q2w3e4r5t

User Credentials are :

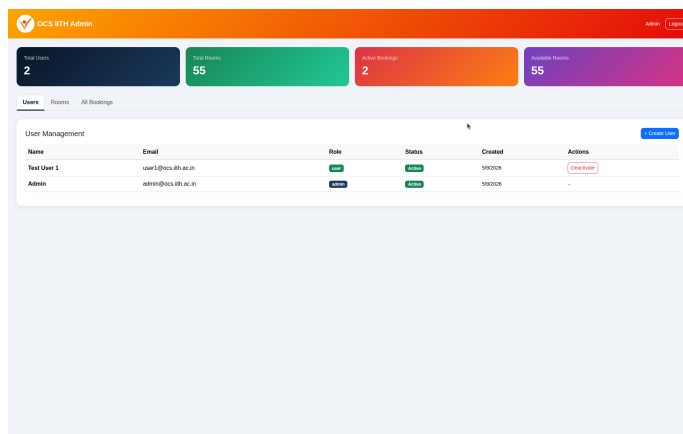
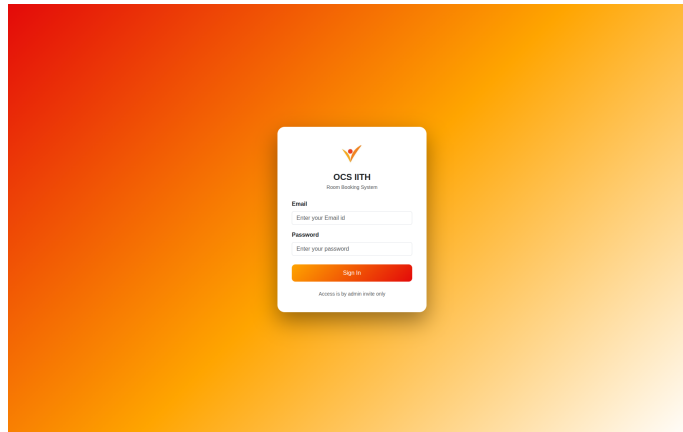
Email : user1@ocs.iith.ac.in

Password : 1q2w3e4r5t

Alternative Way

1. Clone the GitHub repository
2. Open `index.html` with VS Code Live Server
3. Login with the admin credentials
4. The Supabase database is already set up and live, no extra setup needed

8 Screenshots



OCS ETH Admin

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Users

Rooms

All Bookings

Room Management

All Rooms

Block	Room	Capacity	Allowed Purposes	Status	Notes	Actions
A-Block	A-AUDITORIUM	55	DA	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 111	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 112	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 114	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 117	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 118	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 119	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 220	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 221	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 320	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-LH-1	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-LH-2	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
BT-EM-Block	BT-EM-009	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
BT-EM-Block	BT-EM-010	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
BT-EM-Block	BT-EM-114	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>

OCS ETH Admin

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Users

Rooms

All Bookings

All Bookings

2024-05-09

2024-05-09

2024-05-09

Date	Time	Room	Purpose	Booked By	Participants	Status	Action
2024-05-09	21:00 - 21:30	MESE-LH-1 MESE-Block	DA	Test User 1	25	Booked	
2024-05-09	20:30 - 21:30	MA-114 MA-Block	DA	Test User 1	25	Confirmed	<button>Cancel</button>
2024-05-09	20:30 - 21:00	A-Class Room 114 A-Block	DA	Test User 1	25	Confirmed	<button>Cancel</button>

OCS ETH Admin

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Users

Rooms

All Bookings

Room Management

All Rooms

Add New Room

Block *

Room Name *

e.g. A-Block

e.g. A-LH-1

Capacity *

Status

e.g. 50

Available

Allowed Purposes

☒ DA ☒ Interview ☒ PPT

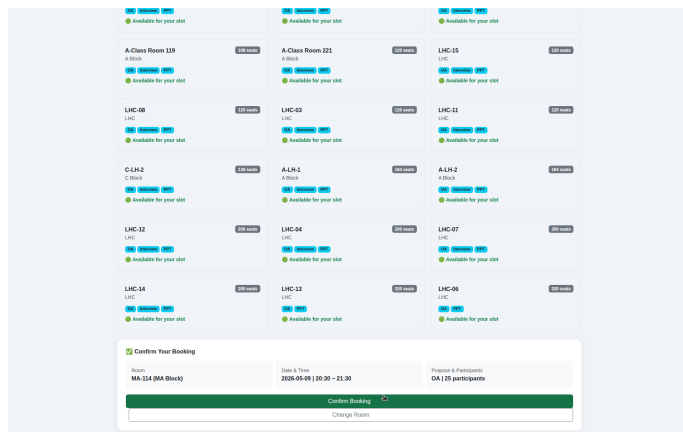
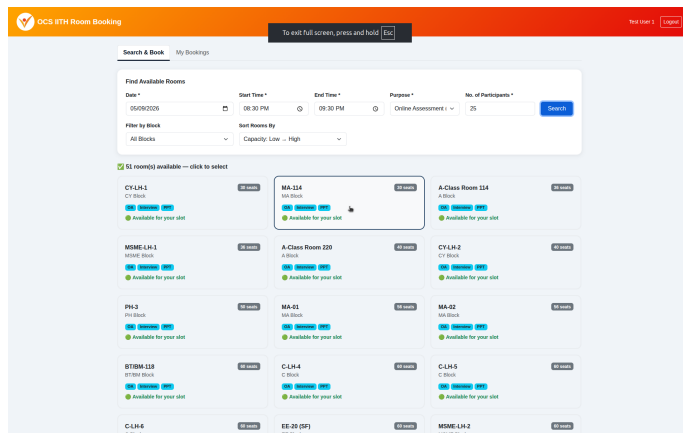
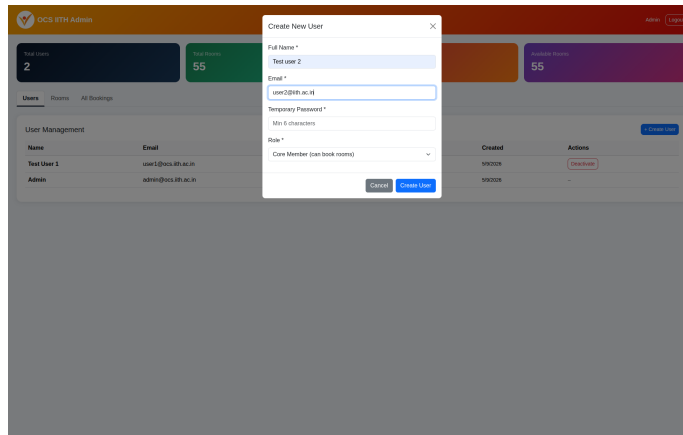
Notes

Any special constraints...

Cancel Add Room

Block	Room	Capacity	Allowed Purposes	Status	Notes	Actions
A-Block	A-AUDITORIUM	55	DA	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 111	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
A-Block	A-Class Room 112	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
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BT-EM-Block	BT-EM-009	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
BT-EM-Block	BT-EM-010	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>
BT-EM-Block	BT-EM-114	25	DA, Interview, PPT	Available		<button>edit</button> <button>delete</button>

5



OCS ITH Room Booking

Two User | Logout

Search & Book

My Bookings

My Bookings

Date	Time	Room	Purpose	Participants	Status	Action
2020-05-09	21:30 - 21:50	MSME-LM 1 General Room	Lab	25	Confirmed	-
2020-05-09	20:30 - 21:30	MA-114 MA-Block	Lab	25	Confirmed	Cancel
2020-05-09	20:30 - 21:30	A-Class Room 114 A-Block	Lab	25	Confirmed	-